

Time Series Analysis

Chapter 13: Speculative bubbles: LPPL models



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Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- **EODHD**: EOD Historical Data (market-data provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- **JLS**: Johansen–Ledoit–Sornette (model)
- **LPPL**: Log-Periodic Power Law
- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **OLS**: Ordinary Least Squares
- **SADF**: Supremum Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SSR**: Sum of Squared Residuals

The question of the chapter

- **Question**: can a speculative bubble be recognised in a price series before it bursts?
  - a **speculative bubble**: a price that rises far above the value justified by the expected cash flows of the asset, sustained by the expectation of selling it later at a higher price
  - a **crash**: a large fall of the price in a short time (here: at least 20% from the peak)
- **Route** of the chapter
  - bubbles in history and the statistical signatures of a bubble
  - rational bubbles and explosive roots: the right-tailed unit-root tests of Phillips, Shi and Yu (link with Chapter 3)
  - the LPPL model of Johansen, Ledoit and Sornette: super-exponential growth, log-periodic oscillations, the critical time
  - estimation, confidence from many windows and the evaluation of crash predictions

Self-study guide

- This chapter is for **self-study**
  - there is no seminar
  - each section ends with a recap
  - read a section, then redo its worked example on paper before you look at the solution
  - after each chart, read the interpretation slide and compare it with your own reading of the chart
- Run the lecture notebook: every chart and number of the slides is produced there
  - change one setting (the window, the filter, the threshold) and see how the conclusion changes
- Check yourself with the self-assessment at the end and with the quiz of Chapter 13 on the course site
- Prerequisites: log returns and log prices (Chapter 1), the Dickey–Fuller test (Chapter 3), OLS regression

Learning outcomes

- Describe a bubble by its fundamental value and by its statistical signatures
- Explain why a rational bubble must grow at an explosive rate and test for explosive roots with SADF and GSADF
- Write the LPPL equation and interpret each of its seven parameters
- Estimate LPPL with the two-step method of Filimonov and Sornette and check the filter conditions
- Build a confidence indicator from many windows and evaluate its alarms honestly, with false alarms and base rates

Reading and tools

- Textbook background: Huang and Petukhina (2022) (unit roots and nonlinear least squares, the tools of this chapter)
  - the LPPL model: Johansen, Ledoit and Sornette (2000); Sornette (2017); calibration: Filimonov and Sornette (2013)
  - explosive roots: Phillips, Wu and Yu (2011); Phillips, Shi and Yu (2015a); survey: Phillips and Shi (2020)
  - history of bubbles: Kindleberger and Aliber (2005); Shiller (2015); Garber (1990)
- Python Quantlets of this chapter: Quantlets/Ch_13
  - the SADF/GSADF statistics and the LPPL calibration are written out in the code (no special package)
- Lecture notebook: open in Google Colab
- Data: daily closes of the S&P 500, Nasdaq 100, BET, Shanghai Composite and Bitcoin (EODHD)

Speculative bubbles, 1636–2021



Flora's wagon of fools, c. 1637

Painting: Hendrik Gerritsz Pot (c. 1637); public domain; Wikimedia Commons

- **Tulip mania**, Netherlands, 1636–1637
  - contracts on rare bulbs traded at extreme prices, then collapsed in February 1637
  - Garber (1990): part of the story is a myth; rare bulbs were expensive for good reasons, common bulbs show the mania
- **South Sea** (1720), **Wall Street** (1929), **Japan** (1989), **dot-com** (2000), **housing** (2007), **crypto** (2017, 2021)
- The common pattern Kindleberger and Aliber (2005): a new story, easy credit, rising prices that attract new buyers, then panic

The crashes of 1929 and 1987



Crowd outside the New York Stock Exchange, 29 October 1929

Photo: US government (1929); public domain; Wikimedia Commons



Amsterdam Stock Exchange, 21 October 1987

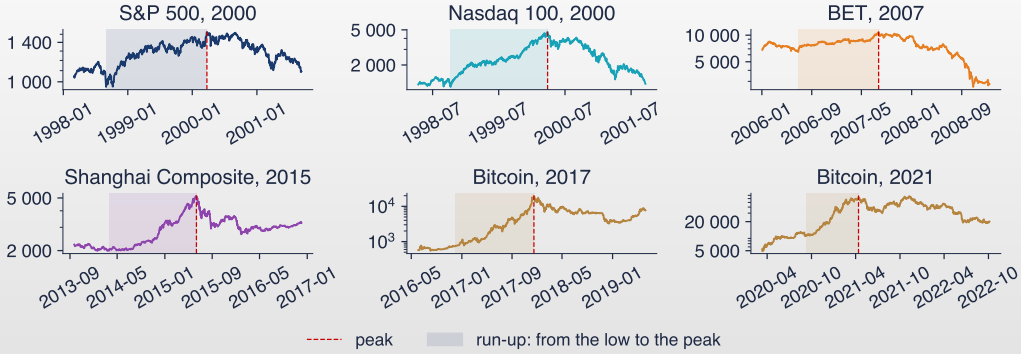
Photo: Bart Molendijk / Anefo, Nationaal Archief (1987); CC0; Wikimedia Commons

- **October 1929**
  - the end of the boom of the 1920s and the start of the Great Depression
- **19 October 1987** (Black Monday)
  - the largest one-day fall of the Dow Jones index, without any major news that day
  - Sornette, Johansen and Bouchaud (1996) found accelerating, log-periodic oscillations in the years before 1987; this observation started the LPPL literature
  - the daily data of this course start in 1990: we study 1987 through the literature and later bubbles with our own data
- A crash without news suggests an **endogenous** cause: the market itself became fragile

Fundamental value and bubble

- **Fundamental value** F_t
  - the present value of the expected future cash flows (dividends D), discounted at the rate r
  - $F_t = \sum_{k=1}^{\infty} \frac{E_t[D_{t+k}]}{(1+r)^k}$
  - D_{t+k}: the dividend paid k periods from now; $E_t[\cdot]$: the expectation given the information available at t; r: the discount rate per period
- **Bubble component**
  - $B_t = P_t - F_t$, the part of the price P_t not explained by fundamentals
  - F_t is not observed: we can only test the **implications** of a bubble in the price path
- Two views
  - efficient markets: prices reflect information; "bubbles" are visible only with hindsight Fama (2014)
  - behavioural finance: herding and extrapolation push prices away from value Shiller (2015); large run-ups do raise the probability of a crash Greenwood, Shleifer and You (2019)

Six run-ups and crashes in the course data



⊡ Daily closes, log scale, from six months before the low of the run-up to eighteen months after the peak; shaded: the run-up from the low to the peak

TSA_ch13_episodes

Interpreting the six episodes

Episode	Low	Peak	Rise	Growth p.a.	Fall in 1 year
S&P 500, 2000	31 August 1998	24 March 2000	60%	30%	27%
Nasdaq 100, 2000	8 October 1998	27 March 2000	317%	97%	66%
BET, 2007	27 June 2006	24 July 2007	68%	48%	50%
Shanghai Composite, 2015	20 March 2014	12 June 2015	159%	77%	49%
Bitcoin, 2017	1 December 2016	16 December 2017	2476%	312%	83%
Bitcoin, 2021	8 September 2020	13 April 2021	527%	309%	53%

- Growth p.a.: $\ln(P_{\mathbf{peak}}/P_{\mathbf{low}})$ divided by the length of the run-up in years; fall: the lowest close in the year after the peak, relative to the peak
- Bitcoin multiplied its price 26 times in about a year; the S&P 500 rose much less: not every peak is a bubble of the same strength

Recap: Bubbles in history

- A bubble is a price above the fundamental value F_t; F_t is not observed
- Six episodes in our data: run-ups of 60% to 2476%, followed by falls of 27% to 83% within a year
- Signatures: explosive and super-exponential growth, accelerating oscillations, then an end

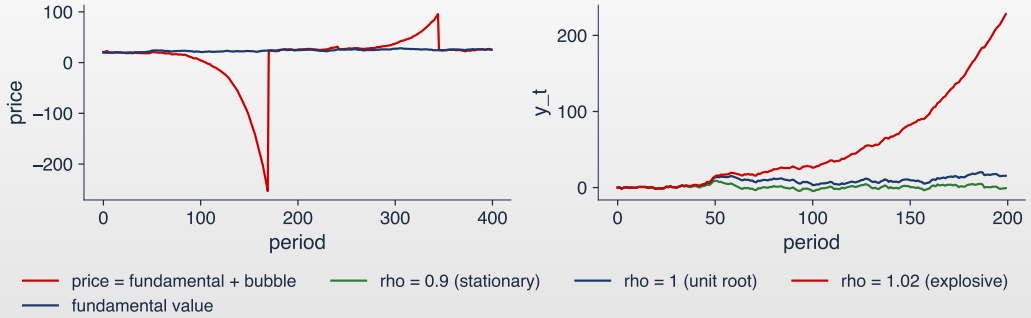
Explosive growth of a rational bubble

- No-arbitrage pricing: $P_t = \dfrac{E_t[P_{t+\mathbf{1}} + D_{t+\mathbf{1}}]}{1+r}$; the fundamental value F_t is one solution
  - in words: today's price is the discounted expected value of tomorrow's price plus tomorrow's dividend
  - every $P_t = F_t + B_t$ with $E_t[B_{t+\mathbf{1}}] = (1+r)\,B_t$ is also a solution
- The bubble must grow, in expectation, at the rate r: $E_t[B_{t+k}] = (1+r)^k B_t$
  - an investor holds an asset above its value only if the overvaluation is expected to grow
  - $1 + r > 1$: the bubble component is an **explosive** process
- Consequence: if dividends are $I(1)$ (Chapter 3), the price is $I(1)$ without a bubble and explosive with one

Worked example: the Blanchard–Watson bubble

- Blanchard and Watson (1982): each period the bubble survives with probability π and then grows to $\frac{1+r}{\pi} B_t$; otherwise it bursts ($B_{t+1} \approx 0$)
  - check: $E_t[B_{t+1}] = \pi \cdot \frac{1+r}{\pi} B_t + (1-\pi) \cdot 0 = (1+r) B_t$
- With $r = 2\%$ and $\pi = 0.98$ per period:
  - while it survives the bubble grows by $1.02/0.98 - 1 = 4.08\%$ per period, more than r: the extra growth pays for the risk of a burst
  - expected lifetime $1/(1-\pi) = 50$ periods; it survives more than 34 periods with probability $1/2$ ($0.98^k = 0.5$)
- The longer the bubble lasts, the faster it must grow and the larger the fall when it bursts

A simulated rational bubble and three AR(1) roots



- Left: fundamental value (random walk) plus a Blanchard–Watson bubble ($r = 2\%$, $\pi = 0.98$), 400 periods; right: $y_t = \rho y_{t-1} + \varepsilon_t$ with the same shocks and $\rho = 0.9$, 1, 1.02

TSA_ch13_rational_bubble

Interpreting the simulation

- The bubble grows, bursts, restarts: in this path it reaches 71 and bursts 2 times after a large size
  - the price inherits the explosive episodes; between them it follows the fundamental value
  - the first bubble of this path is negative and drives the price below zero: free disposal rules this out for a real asset, so a rational bubble must be nonnegative
- AR(1) roots (Chapter 3): $\rho = 0.9$ reverts to the mean, $\rho = 1$ wanders (unit root), $\rho = 1.02$ explodes
  - with $\rho = 1.02$ a shock doubles in about 35 periods and is multiplied by 7.24 after 100
- Idea of the tests: an explosive root ($\rho > 1$) in the log price during a run-up is the statistical trace of a rational bubble

Right-tailed unit-root tests

- The regression of Chapter 3 on a window of the log price $y_t = \ln P_t$: $\Delta y_t = a + \delta\, y_{t-\mathbf{1}} + \varepsilon_t$, with $\delta = \rho - 1$
  - Chapter 3 (Dickey and Fuller (1979)): $H_\mathbf{0}$: $\delta = 0$ against $H_\mathbf{1}$: $\delta < 0$ (stationary), the **left** tail
  - bubbles: $H_\mathbf{0}$: $\delta = 0$ (unit root) against $H_\mathbf{1}$: $\delta > 0$ (explosive), the **right** tail: reject for large t-statistics
- Problem: a bubble occupies only a part of the sample, and the crash after it looks like mean reversion
  - one ADF on the whole sample has almost no power: we must compute the test on many **windows**
- Notation
  - $\Delta y_t = y_t - y_{t-\mathbf{1}}$; a: the intercept; ε_t: the error; ρ: the AR(1) coefficient of y_t
  - $r_\mathbf{1}, r_\mathbf{2} \in [0, 1]$: fractions of the sample; $\mathrm{ADF}_{r_\mathbf{1}}^{r_\mathbf{2}}$: the t-statistic of $\hat{\delta}$ on the window from $r_\mathbf{1} T$ to $r_\mathbf{2} T$; $r_\mathbf{0}$: the smallest window

SADF, GSADF and BSADF (1/2)

- sup: the largest value of the ADF statistic over all the windows considered
  - a single large value, in any window, is enough to signal an explosive episode
- **SADF** Phillips, Wu and Yu (2011)
  - start fixed at 0, the end r_2 moves forward
  - SADF $= \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}$
  - good for one bubble; after a first crash it loses power
- **GSADF** Phillips, Shi and Yu (2015a)
  - both the start r_1 and the end r_2 move
  - GSADF $= \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$
  - detects several bubbles in one sample; smallest window $r_0 = 0.01 + 1.8/\sqrt{T}$

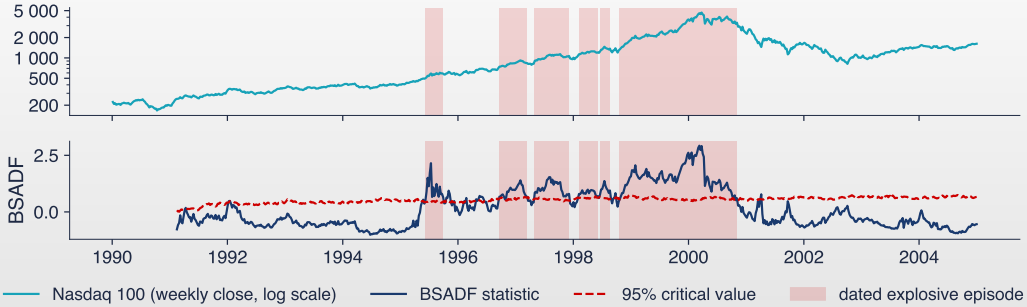
SADF, GSADF and BSADF (2/2)

- **BSADF** (date-stamping)
  - one statistic for each date r_2, using only data up to r_2
  - $\text{BSADF}(r_2) = \sup_{r_1} \text{ADF}_{r_1}^{r_2}$: the largest statistic over all starts r_1 of windows ending at r_2
  - an episode starts when BSADF crosses its critical value and lasts at least $\log T$ observations Phillips, Shi and Yu (2015b) (log: the natural logarithm)
- The distributions are not standard: critical values by Monte Carlo
  - simulated under a random walk with a weak drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$
  - the drift T^{-1} is very small: a unit root, not an explosive root, under H_0
- Reading: an ADF, SADF or GSADF statistic above its 95% critical value rejects the unit root in favour of an explosive root

Worked example: one window of the Nasdaq 100

- Weekly log closes of the Nasdaq 100, $T = 782$ weeks; $r_0 = 0.074$, i.e. a smallest window of 58 weeks
  - end of the window: 24 March 2000, the week of the peak; among all start points the largest statistic comes from the start 17 January 1992
- OLS on this window ($n = 427$ weekly changes): $\widehat{\Delta y_t} = -0.0336 + 0.0060\, y_{t-1}$
  - $t = \hat{\delta}/\mathrm{SE}(\hat{\delta}) = 0.006030/0.002072 = 2.91$
  - so BSADF at the peak is 2.91, above its 95% critical value of 0.54
- Reading: $\hat{\delta} > 0$ means $\hat{\rho} = 1 + \hat{\delta} > 1$: the higher the price, the larger the next weekly rise, the mark of an explosive phase

Explosive episodes of the Nasdaq 100, 1990–2004



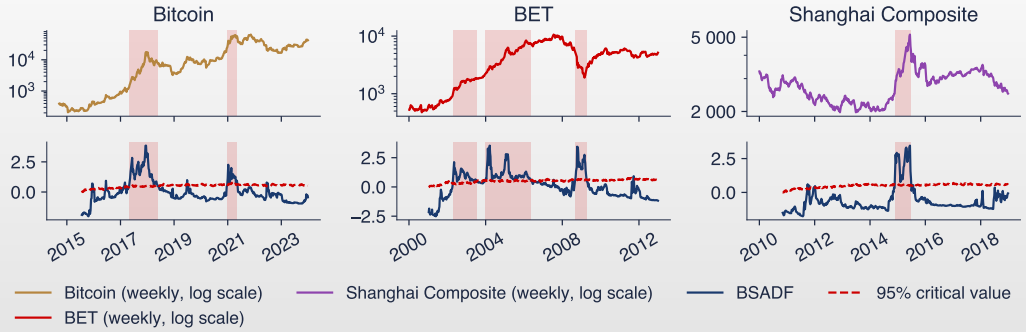
- Top: weekly close, log scale; bottom: BSADF and its 95% critical values (Monte Carlo, 1000 random walks); shaded: dated episodes (at least 7 weeks above the critical value)

TSA_ch13_explosive_roots

Interpreting the Nasdaq 100 tests

- Whole-sample ADF -1.27 (95% critical value -0.13): no evidence
  - SADF 2.72 (critical value 1.48) and GSADF 2.92 (critical value 2.28): explosive behaviour
  - the crash of 2000–2002 hides the bubble from a single test on the whole sample
- BSADF dates 6 episodes; the longest runs from 23 October 1998 to 3 November 2000 (107 weeks)
  - short explosive bursts already from June 1995: the 1990s boom came in waves, as Phillips, Wu and Yu (2011) found
- The end date is late: the statistic stays high for months after the peak; the test dates exuberance, it does not announce the crash

Bitcoin, BET and Shanghai: dated episodes



⊡ Weekly log closes; top: price with the dated episodes; bottom: BSADF and its 95% critical values; Bitcoin 2014–2023, BET 2000–2012, Shanghai Composite 2010–2018

TSA_ch13_explosive_roots

Interpreting the three markets

- GSADF: Bitcoin 3.85, BET 3.53, Shanghai 3.46; above their 95% critical values (2.18, 2.29, 2.15)
  - Bitcoin: April 2017 – May 2018 and January 2021 – May 2021; Shanghai: November 2014 – June 2015
- BET: the explosive phase is December 2003 – May 2006, not the run-up to July 2007, which was strong but not explosive; an earlier episode is dated April 2002 – July 2003
  - the BET flag of September 2008 – April 2009 is the collapse: an **accelerating fall** also gives $\hat{\delta} > 0$; always read the dates against the price chart
- The Bitcoin 2017 episode lasts until May 2018, again well after the peak

Limits of the explosive-root tests

- **Fundamentals**
  - an explosive price is a bubble only if the fundamentals are not explosive
  - better: test the price–dividend ratio (Phillips, Shi and Yu (2015a) for the S&P 500); for Bitcoin there are no dividends at all
- **Volatility**
  - changing volatility distorts the size of the test
  - a wild bootstrap gives robust critical values Phillips and Shi (2020)
- **Timing**
  - BSADF uses only past data, so it can run in real time
  - but it signals a bubble that is already under way and says nothing about when it ends
  - the LPPL model of the next section tries to answer exactly this: when?

Recap: Rational bubbles and explosive roots

- A rational bubble grows in expectation at the rate r: $E_t[B_{t+1}] = (1 + r)B_t$, an explosive process
- Right-tailed ADF: H_1: $\delta > 0$; SADF and GSADF take the supremum over windows; BSADF dates the episodes
- Nasdaq 100, Bitcoin, BET and Shanghai all show explosive episodes; the tests date the run-up, not the crash

Crashes as critical points



Didier Sornette (born 1957)

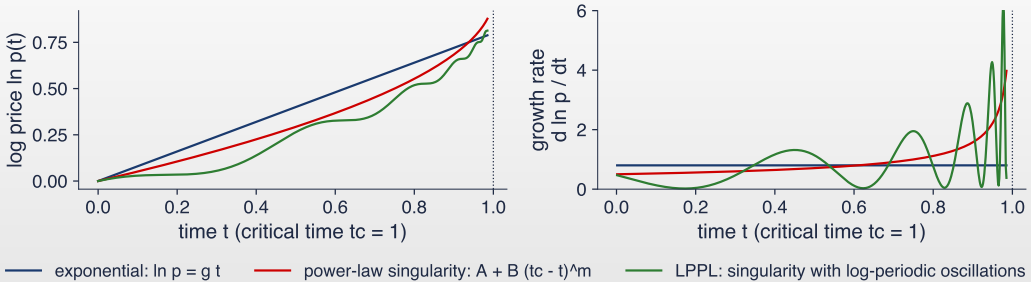
Photo: Didier Sornette (2012); CC BY-SA 3.0 DE; Wikimedia Commons

- Physicist, ETH Zürich; with Anders Johansen and Olivier Ledoit he proposed the **JLS model** Johansen, Ledoit and Sornette (2000)
- Idea: traders imitate their neighbours; imitation creates positive feedback
  - rising prices attract buyers, who push prices higher: growth accelerates
  - the market becomes more and more fragile, like a physical system near a **critical point** (water near boiling)
- The **critical time** t_c
  - the most probable moment of the end of the bubble
  - a crash is likely near t_c but not certain: the bubble may also end with a slow decline

Super-exponential growth

- **Exponential** growth
  - constant growth rate, $\ln P(t) = a + g\,t$
  - $d \ln P/dt = g$
  - a: the initial log price; g: the constant growth rate; $d \ln P/dt$: the growth rate of the price at time t
  - a straight line on a log scale
- **Super-exponential** growth
  - the growth rate increases with time
  - on a log scale the curve bends upward
  - power-law singularity: $\ln P(t) = A + B\,(t_c - t)^m$, with $B < 0$ and $0 < m < 1$
  - t_c: the critical time; $t_c - t$: the time left until it; A: the log price reached at t_c; B: the size of the growth; m: the exponent
  - growth rate $d \ln P/dt = -Bm\,(t_c - t)^{m-\mathbf{1}} \to \infty$ as $t \to t_c$, while $\ln P(t_c) = A$ stays finite
- Such growth cannot last: it must end at or before t_c, a **finite-time singularity**

Exponential and super-exponential growth



- Left: log price; right: its growth rate $d \ln P/dt$; exponential ($g = 0.8$), power-law singularity ($A = 1$, $B = -1$, $m = 0.5$, $t_c = 1$) and the same with log-periodic oscillations

TSA_ch13_lppl_model

Interpreting the growth paths

- All three start with similar growth rates: early in a bubble the paths are hard to tell apart
- The exponential rate stays at 0.8; the singular rate rises from 0.50 to 0.70 halfway and to 3.98 just before t_c
- The oscillations ride on the accelerating trend and become faster near t_c: the next slides explain why

From a crash hazard to the LPPL equation (1/2)

- JLS: the relative price change before the crash has three parts
  - $dP/P = \mu(t)\,dt + \sigma\,dW - \kappa\,dj$
  - $\mu(t)\,dt$: the expected growth over a short interval dt; $\mu(t)$: the drift, which may change in time
  - $\sigma\,dW$: the ordinary noise; W: a Brownian motion (a continuous random walk); σ: the volatility
  - $\kappa\,dj$: the crash; j jumps from 0 to 1 at the crash, which removes a fraction κ of the price
- **Hazard rate** $h(t)$
  - the probability per unit of time that the crash happens now, given that it has not happened yet

From a crash hazard to the LPPL equation (2/2)

- No arbitrage: $E[dP] = 0 \Rightarrow \mu(t) = \kappa\, h(t)$
  - $E[dj] = h(t)\, dt$, so the expected change is zero only if the drift pays for the expected crash loss
  - the higher the hazard, the faster the price must grow: risk and growth increase together
- Imitation near a critical point gives a hazard that rises towards t_c with oscillations
  - $h(t) \approx \alpha(t_c - t)^{m-\mathbf{1}}[1 + \beta\cos(\omega \ln(t_c - t) - \phi')]$
  - $\alpha > 0$: the level of the hazard; $(t_c - t)^{m-\mathbf{1}}$ grows without bound as $t \to t_c$, since $m < 1$
  - β: the relative size of the oscillations; ω: their log-frequency; ϕ': their phase
- Integrating $\mu(t) = \kappa h(t)$ over time gives the log price on the next slide

The LPPL equation

- **Log-periodic power law** (LPPL), for $t < t_c$
  - $\ln P(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \phi\big)$

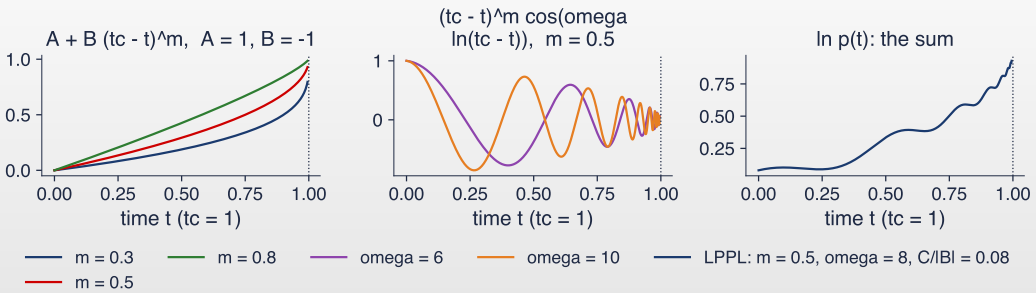
Parameter	Meaning	Usual range
t_c	critical time: most probable end of the bubble	after the last observation
A	log price at t_c (if the bubble reached t_c)	$A > 0$
B	size of the power-law growth	$B < 0$ (rising price)
m	exponent: how fast the growth accelerates	$0 < m < 1$
C	relative size of the oscillations	$\lvert C\rvert < \lvert B\rvert$
ω	angular log-frequency of the oscillations	$2 \leq \omega \leq 25$
ϕ	phase of the oscillations	$[0, 2\pi)$

- Seven parameters: three of them (t_c, m, ω) enter nonlinearly; ϕ disappears in the estimation (Section 5)

Log-periodic oscillations

- $\cos(\omega \ln(t_c - t) - \phi)$ is periodic in $\ln(t_c - t)$, not in t: one full cycle each time $\ln(t_c - t)$ changes by $2\pi/\omega$
  - successive peaks t_n satisfy $\dfrac{t_c - t_n}{t_c - t_{n+1}} = \lambda = e^{2\pi/\omega}$: the time left to t_c shrinks by the same factor at each cycle
- **Discrete scale invariance**
  - the pattern looks the same when time to t_c is rescaled by λ (as in a hierarchy of traders, groups, institutions)
- Worked example: with $\omega = 8$, $\lambda = e^{2\pi/8} = 2.19$
  - if a correction happens 200 days before t_c, the next one comes about $200/2.19 \approx 91$ days before t_c, then about 42, then 19
  - $\omega = 6.28$ gives $\lambda = 2.72$; $\omega = 10$ gives $\lambda = 1.87$; the empirical literature reports λ near 2 Sornette (2017)

The pieces of the LPPL equation



- Left: $A + B(t_c - t)^m$ for $m = 0.3, 0.5, 0.8$; middle: the oscillation $(t_c - t)^m \cos(\omega \ln(t_c - t))$ for $\omega = 6$ and 10; right: the full LPPL path ($m = 0.5$, $\omega = 8$, $C/|B| = 0.08$), which fails the damping condition: $m|B|/(\omega|C|) = 0.78 < 1$; $t_c = 1$

TSA_ch13_lppl_model

Interpreting the components

- A smaller m concentrates the growth close to t_c: the curve is flat for a long time, then shoots up
  - m close to 1 is almost a straight line: hardly a bubble; m close to 0 is a jump at t_c
- The oscillation term shrinks with $(t_c - t)^m$ and speeds up: the cycles get shorter by the factor λ
  - a larger ω gives more cycles before t_c
- Real run-ups (Section 5) show this mixture: an accelerating trend with corrections that come more and more often

Worked example: an LPPL value by hand

- Parameters: $A = 8$, $B = -1.2$, $C = 0.08$, $m = 0.5$, $\omega = 8$, $\phi = 0$, $t_c = 2$ (years); compute $\ln P(t)$
  - $t = 1$: $t_c - t = 1$, $(t_c - t)^m = 1$, $\ln 1 = 0$, $\cos 0 = 1$: $\ln P = 8 - 1.2 + 0.08 = 6.880$, $P = 973$
  - $t = 1.75$: $(0.25)^{\mathbf{0.5}} = 0.500$, $\ln 0.25 = -1.386$, $\cos(8 \cdot (-1.386)) = 0.095$: $\ln P = 8 - 1.2 \cdot 0.500 + 0.08 \cdot 0.500 \cdot 0.095 = 7.404$, $P = 1642$
  - $t = 1.99$: $(0.01)^{\mathbf{0.5}} = 0.100$, $\cos(8 \cdot (-4.605)) = 0.654$: $\ln P = 7.885$, $P = 2658$
- The price rises from 973 to 2658 as t goes from 1 to 1.99 and would reach $e^A = 2981$ at t_c; most of the rise comes in the last months

Parameter constraints: the filter

- A fit is **qualified** only if it describes a credible bubble (Shu and Zhu (2020), following Sornette et al. (2015)):
  - $B < 0$: rising price; $0.01 \leq m \leq 0.99$: accelerating growth with a finite price at t_c
  - $2 \leq \omega \leq 25$: neither too slow nor too fast oscillations; at least 2.5 oscillations in the window, $\frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$
  - $t_2 \leq t_c \leq t_2 + 0.2\,(t_2 - t_1)$: the end is close, relative to the length of the window $[t_1, t_2]$
  - damping $\dfrac{m|B|}{\omega|C|} \geq 1$: the hazard rate stays positive
- Conditions on the residuals
  - every fitted price within 15% of the observed price; the Lomb periodogram Lomb (1976) confirms the log-periodic cycle (level 10%)
  - Lomb periodogram: a periodogram for unevenly spaced points, here of the residuals as a function of $\ln(t_c - t)$
  - the residuals are stationary: the Dickey–Fuller and Phillips–Perron tests of Chapter 3 reject a unit root (level 10%)

Recap: The LPPL model

- Imitation creates a hazard rate that grows towards t_c; no arbitrage turns it into super-exponential growth
- $\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$, seven parameters
- Log-periodic: cycles shrink by $\lambda = e^{2\pi/\omega}$; a fit counts only if it passes the filter

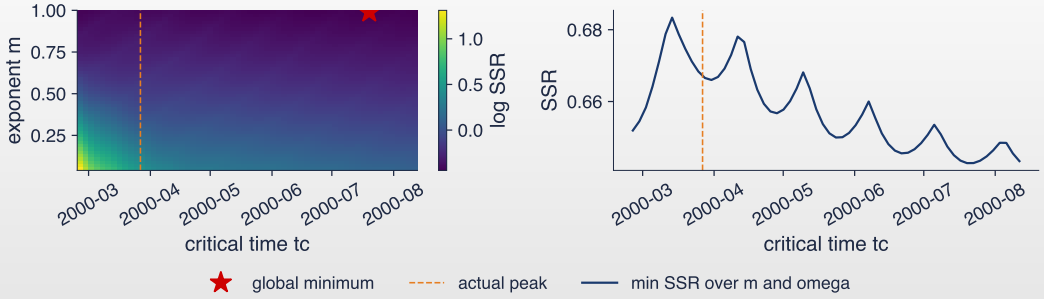
The two-step method of Filimonov and Sornette (1/2)

- Least squares over seven parameters has many local minima; Filimonov and Sornette (2013) reduce the search to three parameters
  - $\min \mathrm{SSR} = \sum_{t=t_1}^{t_2} [\ln P_t - \mathrm{LPPL}(t)]^2$; SSR: the sum of squared residuals over the window $[t_1, t_2]$
  - $\mathrm{LPPL}(t)$: the right-hand side of the LPPL equation
- Expand the cosine, with $\tau = t_c - t$
  - $C \cos(\omega \ln \tau - \phi) = C_1 \cos(\omega \ln \tau) + C_2 \sin(\omega \ln \tau)$, where $C_1 = C \cos \phi$, $C_2 = C \sin \phi$
- The LPPL equation becomes linear in A, B, C_1, C_2
  - $\ln P(t) = A + B\, f_t + C_1\, g_t + C_2\, h_t$
  - $f_t = \tau^m$, $g_t = \tau^m \cos(\omega \ln \tau)$, $h_t = \tau^m \sin(\omega \ln \tau)$: known numbers once t_c, m, ω are fixed

The two-step method of Filimonov and Sornette (2/2)

- **Step 1** (linear)
  - for given (t_c, m, ω), regress $\ln P_t$ on $(1, f_t, g_t, h_t)$ by OLS
  - $(\hat{A}, \hat{B}, \hat{C}_1, \hat{C}_2) = (X'X)^{-1}X'y$; X: the matrix with columns $1, f_t, g_t, h_t$; y: the vector of $\ln P_t$
  - this gives the concentrated cost $\mathrm{SSR}(t_c, m, \omega)$, a function of three parameters only
- **Step 2** (nonlinear)
  - minimise $\mathrm{SSR}(t_c, m, \omega)$ over the search space
  - $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$, $m \in [0, 1]$, $\omega \in [1, 50]$
  - in the code: a grid of $8 \times 8 \times 16$ points, then the Nelder–Mead simplex from the best grid point
- At the end: $C = \sqrt{C_1^2 + C_2^2}$ and $\phi = \mathrm{atan2}(C_2, C_1)$
  - $\mathrm{atan2}(C_2, C_1)$: the angle of the point (C_1, C_2), in $(-\pi, \pi]$

The cost landscape



- Nasdaq 100, window from the low of October 1998 to 30 days before the peak
  - left: $\log \text{SSR}$ over (t_c, m), minimised over $\omega \in [2, 25]$ (135 360 grid points, with the linear step solved at each point)
  - right: the minimum over m and ω as a function of t_c

TSA_ch13_estimation

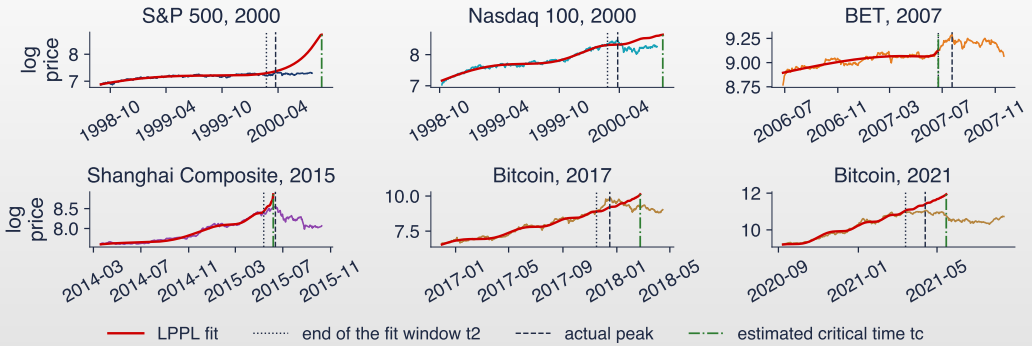
Interpreting the cost landscape

- The profile over t_c has 5 local minima: an optimiser started at a different t_c can end in a different valley
  - the waves come from the oscillation term: shifting t_c shifts the phase of the cycles
- The landscape is flat: the worst and the best t_c differ by only 6.3% in SSR
  - the global minimum is at m near 1 and $t_c = 19$ July 2000, far after the actual peak: the data barely identify t_c
- Lesson: one LPPL fit is never enough; we need many windows and the filter (Section 6)

Worked example: the Shanghai fit at the end of the window

- Shanghai Composite, window 20 March 2014 – 13 May 2015 (281 days); estimates: $t_c = 2015.428$ (6 June 2015), $m = 0.392$, $\omega = 5.50$
  - step 1 gives $\hat{A} = 8.893$, $\hat{B} = -1.270$, $\hat{C}_1 = 0.020$, $\hat{C}_2 = 0.074$, so $\hat{C} = 0.076$
- At $t_2 = 2015.361$ (years): $\tau = t_c - t_2 = 0.067$, $\tau^m = 0.347$, $\ln \tau = -2.702$
  - $\cos(\omega \ln \tau) = -0.656$, $\sin(\omega \ln \tau) = -0.754$
  - $\widehat{\ln P} = 8.893 + (-0.440) + (-0.024) = 8.429$; observed $\ln P = 8.384$
- In levels: fitted 4579 against observed 4376, an error of 4.7%; the three terms are the level A, the power-law growth and the oscillation

LPPL fits 30 days before the peak



- Window: from the low before the run-up to 30 calendar days before the peak; the fitted path is extended to the estimated t_c; data after the window are shown only for comparison

TSA_ch13_estimation

Interpreting the six fits

Episode	t_2	$\hat{t}_c$	Days to peak	$\hat{m}$	$\hat{\omega}$	Param.	All
S&P 500, 2000	23 February 2000	21 August 2000	+150	1.00	1.0	no	no
Nasdaq 100, 2000	25 February 2000	11 August 2000	+137	1.00	5.9	no	no
BET, 2007	22 June 2007	22 June 2007	−32	0.69	1.0	no	no
Shanghai Composite, 2015	13 May 2015	6 June 2015	−6	0.39	5.5	yes	yes
Bitcoin, 2017	16 November 2017	22 February 2018	+68	0.71	14.1	no	no
Bitcoin, 2021	14 March 2021	15 May 2021	+32	0.82	17.4	no	no

- Days to peak: $\hat{t}_c$ minus the actual peak; Param.: the fit passes the parameter conditions; All: it also passes the residual conditions
- Only Shanghai passes the whole filter, with $\hat{t}_c$ within a week of the peak; S&P 500 and Nasdaq 100 hit the boundary $m = 1$: no super-exponential growth in these windows
- Bitcoin fails the 15% error condition (its daily swings are too large for one smooth path) and the t_c condition ($\hat{t}_c$ too far after t_2)

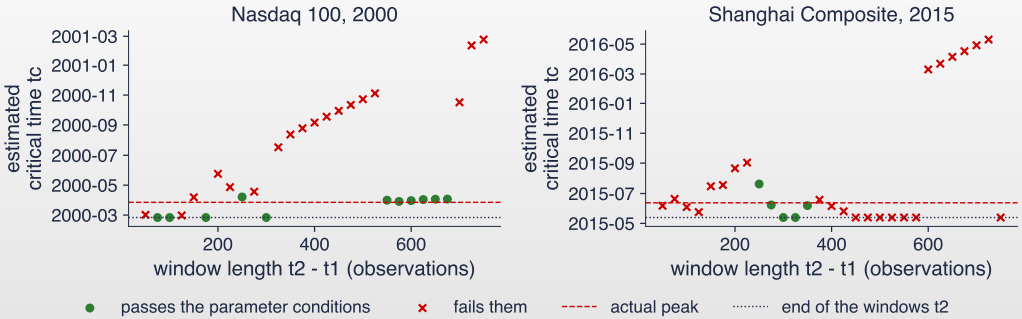
Recap: Estimation

- Four linear parameters by OLS, three nonlinear ones (t_c, m, ω) by search: the method of Filimonov and Sornette (2013)
- The cost surface is rugged and flat in t_c: the critical time is weakly identified
- One window, one fit: only one of six episodes gives a fully qualified fit 30 days before the peak

The LPPLS confidence indicator

- So far t_1 was the low before the run-up, a date known only **after** the bubble; in real time the start is unknown
  - remedy: for each end date t_2 (today), fit LPPL on 29 windows $[t_2 - L, t_2]$, $L = 750, 725, \ldots, 50$ trading days Shu and Zhu (2020)
- Indicator = share of the windows ending at t_2 whose fit is qualified Sornette et al. (2015); 0: no window sees a bubble, 1: all do
  - LPPLS (LPPL singularity) is the name used for this indicator; the equation is the same
- Two versions in this chapter
  - **parameter conditions**: B, m, ω, t_c, oscillations, damping
  - **all conditions**: also the 15% error, the Lomb test and stationary residuals (Shu and Zhu (2020))
- Computed only with data up to t_2: a real-time indicator; an alarm when it exceeds a threshold fixed in advance (here 0.2)

Estimated critical time across estimation windows



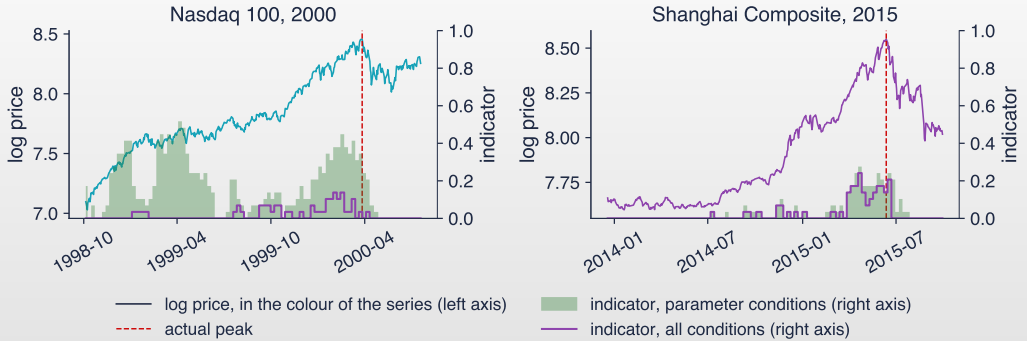
- Estimated t_c for each of the 29 windows ending 30 days before the peak; green dots: the fit passes the parameter conditions; red crosses: it fails them

TSA_ch13_confidence

Interpreting the windows

- Nasdaq 100 (end 25 February 2000): 11 of 29 windows pass the parameter conditions, 1 the whole filter
  - qualified $\hat{t}_c$: median 30 March 2000, 10%–90% range 25 February 2000 – 2 April 2000; actual peak 27 March 2000
- Shanghai (end 13 May 2015): 5 of 29 pass the parameter conditions; median $\hat{t}_c$ 6 June 2015, actual peak 12 June 2015
  - the condition on m fails most often: only 41% of the Shanghai windows have $0.01 \leq m \leq 0.99$
- Short windows put t_c right after t_2; long windows spread it out: the window length is a hidden modelling choice

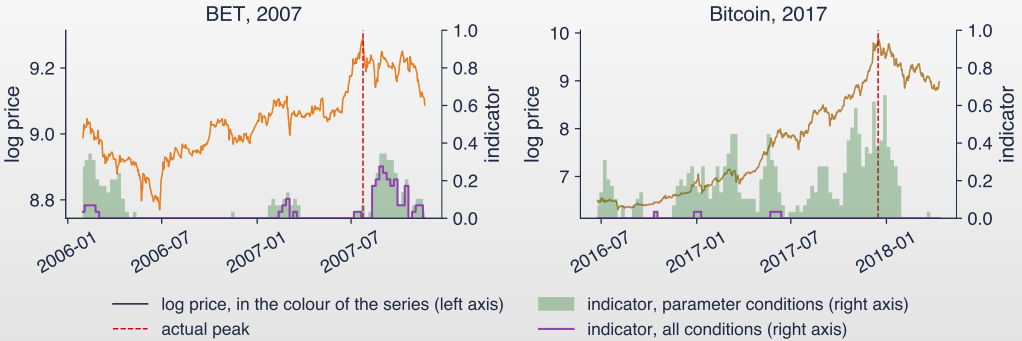
The indicator around four peaks (1/2)



- Log price (left axis) and the indicator (right axis) every 5 trading days, from 18 months before to 4 months after the peak
  - green area: parameter conditions; purple line: all conditions

TSA_ch13_confidence

The indicator around four peaks (2/2)



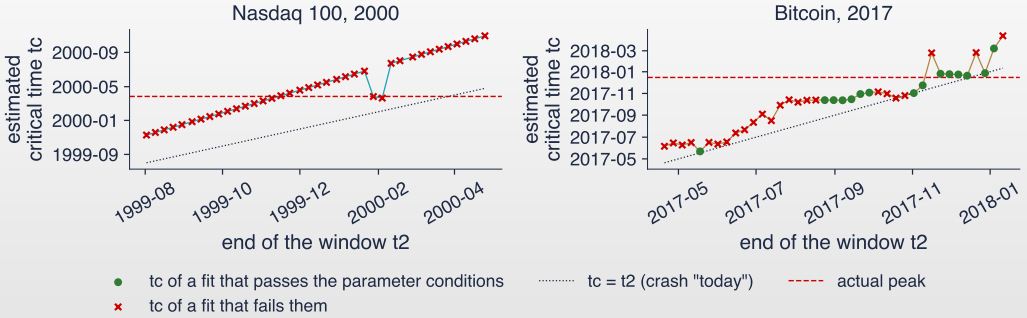
- The same layout and colours as in (1/2), for the BET (2007) and Bitcoin (2017); Bitcoin: one value every 7 days

TSA_ch13_confidence

Interpreting the indicator

- Nasdaq 100: the parameter version reaches 0.52 on 7 April 1999, a year before the peak, and rises again in the last months; the full version stays below 0.14
  - Shanghai: both versions rise in the two months before the peak (maximum on 22 April 2015: 0.24 with the full filter, 0.28 with the parameter conditions only)
- BET: the strongest full-filter signal, 0.28, comes on 28 August 2007, after the peak: the fits then describe the last part of the run-up, not the future
  - Bitcoin 2017: the parameter version peaks at 0.66 on 29 December 2017; the full version is almost always 0 (the 15% error condition)
- The indicator does light up in bubbles, but early, late and with gaps; the choice of filter changes the picture

Instability of the estimated critical time



- Window start fixed at the low before the run-up; the end t_2 moves from 8 months before to 1 month after the peak (weekly); each point: the $\hat{t}_c$ of the fit ending at t_2

TSA_ch13_confidence

Interpreting the moving critical time

- Nasdaq 100: over 38 end dates $\hat{t}_c$ ranges from 9 November 1999 to 30 October 2000 (356 days); none of these fits passes the parameter conditions
  - Bitcoin 2017: range 21 May 2017 – 12 April 2018; 38% of the fits pass; median error 65 days
- Bitcoin: 4 of the 39 fits put $\hat{t}_c$ exactly at t_2 and 11 within 20 days of it: the fit then says "the end is now"; Nasdaq 100: $\hat{t}_c$ mostly sits at the upper bound of the search space
  - a forecast that keeps moving with the data is a description of the past, not a prediction
- Picking, after the crash, the one window whose $\hat{t}_c$ was right is **look-ahead bias**

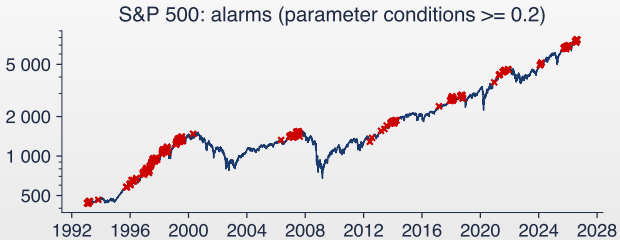
Recap: Estimation over many windows

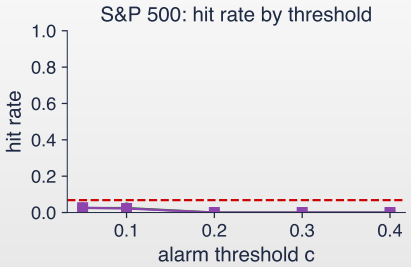
- The start of a bubble is unknown: fit 29 windows for each end date
- Indicator = share of qualified windows; it depends strongly on the filter
- $\hat{t}_c$ moves with the end of the data: report its spread over windows, never one value

Common errors in evaluating crash predictions

- **Look-ahead bias**
  - using information from after t_2
  - choosing t_1 at the low of the run-up, the episodes after the crash, or the filter after seeing the results
- **Selection**
  - studying only the bubbles that burst
  - the run-ups that did not end in a crash are forgotten
- **False alarms and base rates**
  - an alarm is useful only if a crash is **more likely after an alarm** than on an ordinary day
  - the comparison: $P(\text{crash} \mid \text{alarm})$ against $P(\text{crash})$, the unconditional frequency
- A fair test fixes everything in advance (windows, filter, threshold, the definition of a crash) and runs over the whole sample

Alarms over the whole sample (1/2)



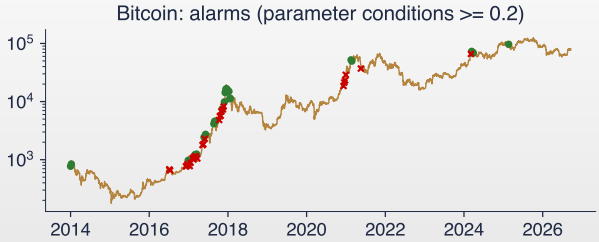


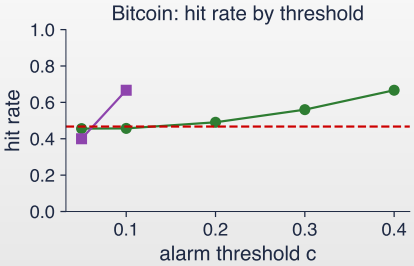
S&P 500 · alarm followed by a fall of 20% or more · false alarm · hit rate, parameter conditions · hit rate, all conditions · unconditional frequency

- S&P 500: indicator every 5 trading days, January 1993 – September 2026
  - a crash: a fall of at least 20% within 182 days after the alarm; right: hit rate by threshold

TSA_ch13_evaluation

Alarms over the whole sample (2/2)





Bitcoin
alarm followed by a fall of 20% or more
false alarm
hit rate, parameter conditions
hit rate, all conditions
unconditional frequency

- Bitcoin: indicator every 7 days, January 2014 – September 2026
  - hit rate: the share of the alarm dates followed by a crash; dashed line: the unconditional frequency

TSA_ch13_evaluation

Interpreting the evaluation

- S&P 500: a fall of 20% within six months follows 7% of all dates, but 0% of the alarm dates (threshold 0.2, parameter conditions)
  - 20 alarm clusters, none followed by such a fall within six months; the alarms come in calm, steady uptrends
  - of the 4 drawdowns of 20% or more since January 1993, 2 had an alarm in the six months before their peak (2007 and 2022), but the falls came later
- Bitcoin: base rate 47%, after an alarm 49%: almost no gain; 12 of its 14 drawdowns of 20% had an alarm before the peak
  - falls of 20% are so frequent in Bitcoin that almost any alarm looks "right"; higher thresholds help a little, on very few dates (56% at 0.3, on 4% of the dates)
- The indicator says "this looks like a bubble regime", not "the crash comes within six months"

The literature on crash prediction

- Successes reported by the authors of the method
  - Jiang et al. (2010): the Chinese bubbles of 2007 and 2009 diagnosed in advance; Sornette et al. (2015): the Shanghai 2015 peak announced before it happened
  - Gerlach, Demos and Sornette (2019): Bitcoin's bubbles of 2012–2018 dissected with the confidence indicator
- Critical evaluations
  - Feigenbaum (2001): the log-periodic term adds little once the estimation error is taken into account
  - Brée and Joseph (2013): on many crashes, the LPPL conditions are rarely all met before the crash; parameters are unstable
- Our own numbers point the same way: the indicator reacts to strong run-ups, but its timing is loose and its alarms are often false

Strengths and limits of LPPL

- **Can**
  - describe a run-up with accelerating growth and shrinking corrections in a few interpretable parameters
  - measure "how bubble-like" a market is today, across many windows (the indicator)
- **Cannot**
  - give the date of a crash: t_c marks the end of the regime
  - guarantee a crash: the regime may end with a slow decline
  - tell whether the price is above the fundamental value: that needs fundamentals
- Use it with the explosive-root tests, with fundamentals (valuation ratios) and with risk measures such as VaR 1% (Chapter 5), not alone

Recap: Evaluation of crash predictions

- Fix windows, filter, threshold and the definition of a crash before looking at the outcomes
- Compare the hit rate after alarms with the base rate; count the false alarms
- LPPL describes bubbles well; as a crash-timing tool its record is weak

Possible contribution of AI

- **Code**
  - a first version of the BSADF recursion, of the Filimonov–Sornette two-step fit, of the indicator over many windows
- **Explanation**
  - a second explanation of the hazard-rate argument or of log-periodicity, with your own numbers
- **Exploration**
  - the indicator on many markets and periods, with different filters, under a protocol fixed in advance
- Example prompt
  - `Write Python code that fits the LPPL model to the daily log price of the Shanghai Composite between two dates with the Filimonov-Sornette method (OLS for A, B, C1, C2; grid search plus Nelder-Mead for tc, m, omega), checks the filter conditions of Shu and Zhu (2020) and plots the fit.`

Necessary checks

- Simulate an LPPL path with known parameters plus noise and check that the code recovers t_c, m and ω
- The sign conventions: $B < 0$ for a rising bubble; the test of explosive roots uses the right tail, not the left tail of Chapter 3
- Critical values: SADF and GSADF need their own simulated critical values, not the Dickey–Fuller table
- No look-ahead: every quantity at date t_2 must use only data up to t_2
- A crash "predicted" by an AI answer after the fact is not a prediction; ask for the full list of alarms and false alarms
- Every cited reference: it must exist; check the DOI

Key takeaways

- A bubble is a price above its fundamental value; a rational bubble must grow explosively
- Right-tailed ADF tests (SADF, GSADF, BSADF) detect and date explosive episodes with data up to each date
- LPPL: super-exponential growth with log-periodic oscillations ending at a critical time t_c
- Estimate in two steps (OLS + search over t_c, m, ω), filter the fits, use many windows
- Judge predictions by hit rates against base rates, with all the false alarms: the timing of crashes remains hard

Key formulas

Quantity	Formula
Rational bubble	$P_t = F_t + B_t, \quad E_t[B_{t+1}] = (1+r)B_t$
Right-tailed ADF	$\Delta y_t = a + \delta y_{t-1} + \varepsilon_t, \quad H_1\colon \delta > 0$
GSADF, BSADF	$\sup_{r_2} \sup_{r_1} \mathrm{ADF}_{r_1}^{r_2}, \quad \mathrm{BSADF}(r_2) = \sup_{r_1} \mathrm{ADF}_{r_1}^{r_2}, \quad r_0 = 0.01 + 1.8/\sqrt{T}$
LPPL	$\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$
Linear form	$\ln P(t) = A + Bf_t + C_1 g_t + C_2 h_t, \quad C = \sqrt{C_1^2 + C_2^2}$
Scaling ratio	$\lambda = e^{2\pi/\omega}$
Damping	$m\lvert B\rvert/(\omega\lvert C\rvert) \geq 1$
Indicator	share of qualified windows ending at t_2

Self-assessment (1/2)

- **Question**: in a Blanchard–Watson bubble with $r = 1\%$ and $\pi = 0.95$, by how much does the bubble grow in a period in which it survives?
  - **Answer**: by $1.01/0.95 - 1 \approx 6.3\%$; expected lifetime $1/(1 - 0.95) = 20$ periods
- **Question**: a right-tailed ADF on a window gives $\hat{\delta} = 0.004$ with SE 0.002. Is the window explosive at 5% if the critical value is 1.5?
  - **Answer**: yes: $t = 0.004/0.002 = 2 > 1.5$; we reject the unit root in favour of an explosive root
- **Question**: why do we use the right tail and not the left tail of Chapter 3?
  - **Answer**: the alternative is $\rho > 1$ (explosive), not $\rho < 1$ (stationary); evidence for it is a large positive t

Self-assessment (2/2)

- **Question**: an LPPL fit gives $\omega = 6.28$. By what factor does the time left to t_c shrink between two corrections?
  - **Answer**: $\lambda = e^{2\pi/6.28} \approx e \approx 2.72$
- **Question**: a fit has $m = 1.0$ at the edge of the search space. Is it a qualified bubble fit?
  - **Answer**: no: the filter needs $m \leq 0.99$; $m = 1$ is a linear trend in t, not an accelerating one
- **Question**: an indicator gives alarms before 8 of 10 crashes. Is it a good predictor?
  - **Answer**: not yet: we also need the false alarms and the base rate; if alarms are on almost every day, catching 8 crashes is easy

Project idea

- **Is the BET in a bubble today?** A real-time protocol, fixed before looking at the result
  - BSADF on the weekly BET and BET-TR; the indicator of this chapter on the daily BET
  - the evaluation of Section 7 on the BET since 2000: alarms, false alarms, base rate
- Extensions
  - the price–earnings ratio instead of the price; Bitcoin and Ethereum together
  - compare with the regime-switching models of Chapter 10
- Further reading on the Bucharest Stock Exchange: Pele and Mazurencu-Marinescu (2012)
- Next: Chapter 14, multivariate GARCH models

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